ENEE12014

LECTURE NOTES

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Contents

1 Introduction

1.1 Electricity:

Through this is a steady of moving charges and their interaction until magnetic and electric fields change this. Charges move because a net force have been applied. The force comes from a difference in charge between two points on a circuit.

1.2 SI Units:

$$1\ Coulomb = 6.25 \times 10^{18} e^{-}$$

$$1A = \frac{1Q(Coulomb)}{1t(Seconds)}$$

1.3 Electrical Potential Energy:

The potential energy is closely associated with an object's position. E_{PE} is what all electrical circuits have. To move charges within an electric field it requires an amount of work.

1.4 Voltage:

$$1V = \frac{1J}{C}$$

Defines the potential difference between two points in a conductor.

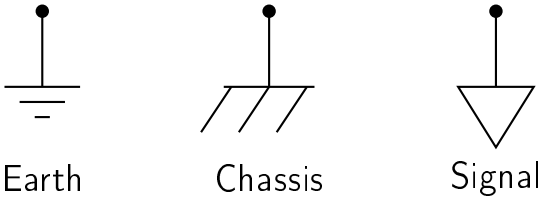
1.5 Elctro Motive Force:

emf is the energy provided by an eneyr source to enable the flow of charges.

1.6 Grounding:

When grounding a circuit we are basically joining it a giant resvior of both positive and negative charge. This is large enough that it can both recieve and donate charge. As such when an object comes into contact with group all net charge is removed becoming something with OV.

Ground Circuit Examples



1.7 Passive Circuit Element:

Conductors are circuit elements that allow the free movement of charge. Super conductors do this with no resistance. Insulators are the opposite of conductors. They prevent the free movement of charge. A semi-conductor has properties of conductors and insulators.

1.8 Resistance:

Resistivity (ρ) is how strongly it resists electrical current.

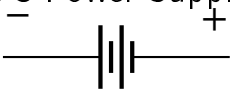
$$R = \frac{\rho L}{A}$$

Where:

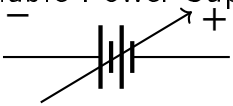
- R: Resistance (Ω)
- ρ: Resistivity ($(\Omega \cdot m)^{-1}$)
- L: Lenght of wire (m)
- A: CSA (m^2)

1.9 Circuit Symbols:

DC Power Supply



Variable Power Supply



AC Power Supply



Inductor



Wire

Junction



Ampmeter



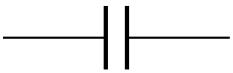
Volt meter



Resistor



Capacitor



Voltage rises across from the negative to positive terminals of the components. In addition to the opposite flow to the amps. Current divider rule:

$$I_1 = I_S \cdot \frac{R_2}{R_1 + R_2}$$

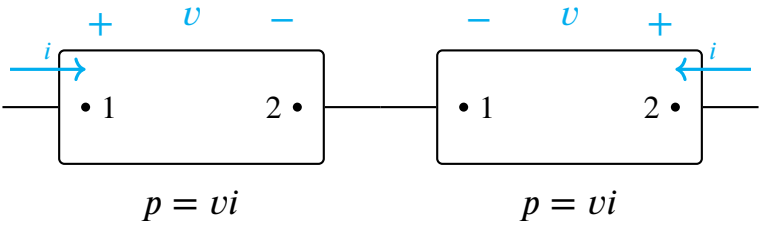
$$I_2 = I_S \cdot \frac{R_1}{R_1 + R_2}$$

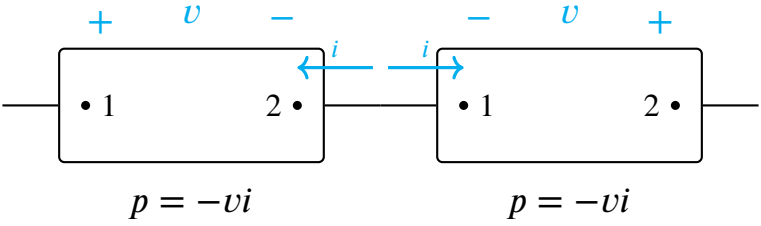
2 Circuit Simplification

2.1 Ideal Power Sources:

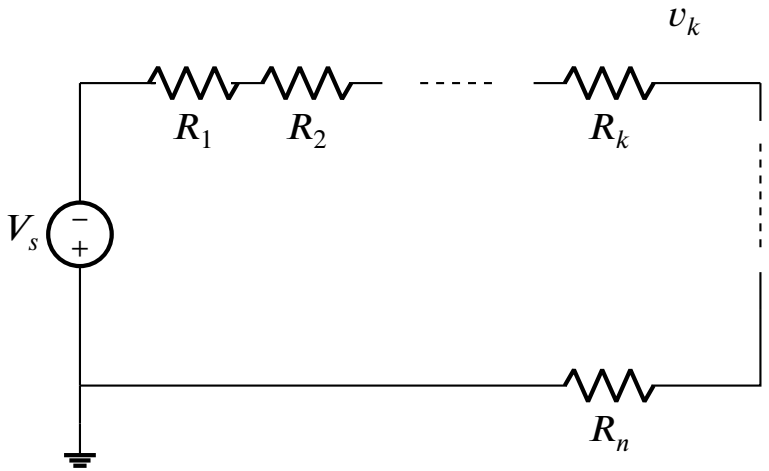
For ideal voltage and current sources, there are independent and dependent sources. The independent source are illustrated by a circle around the indicator and its numerical value is just defined. The dependent source usually rely on a contast multiplied by a either a current or voltage value within the circuit. They are also illustrated through a diamond shape.

2.2 Polarity and Power:





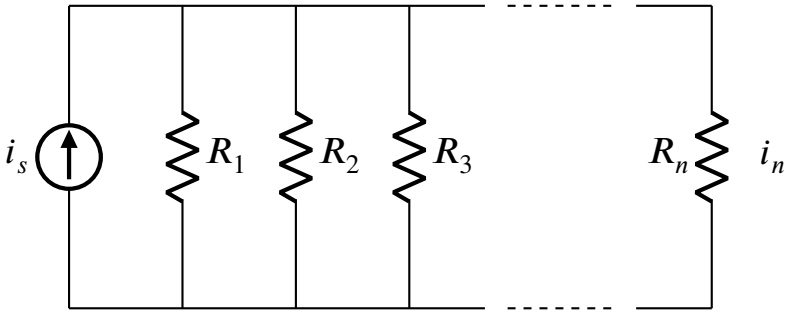
2.3 Voltage Divider Rule:



For this to work the current through all resistors must be equal. The highest voltage drop occurs over the largest resistor.

$$V_k = \frac{R_k}{R_{eq}} \cdot V_S$$

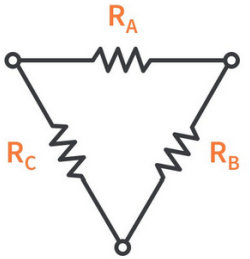
2.4 Current Divider Rule:

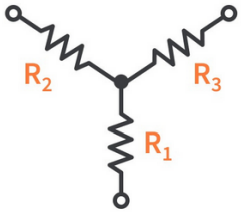


The highest current goes through the smallest resistor.

$$I_k = \frac{I_S}{R_k R_{eq}}$$

2.5 Delta-Wye Resistor Configurations:





$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1} \qquad R_1 = \frac{R_bR_c}{R_a + R_b + R_c}$$

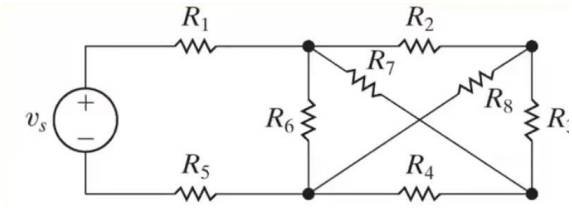
$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2} \qquad R_2 = \frac{R_cR_a}{R_a + R_b + R_c}$$

$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3} \qquad R_3 = \frac{R_aR_b}{R_a + R_b + R_c}$$

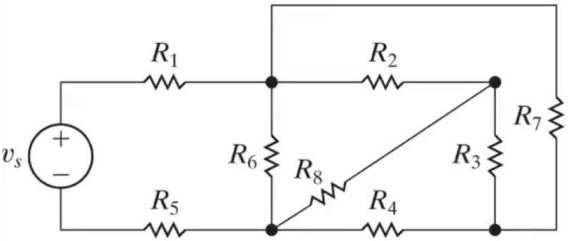
3 Nodal Analysis

3.1 Planar Circuit:

A circuit in which there are no crossing branches that can be drawn on plane. In the image shown below, the top circuit can be redrawn to the bottom one. Turning it into a planar circuit.



(a)



3.2 Essential Points:

Node: A point in a circuit which two or more elements join.
Essential Node: Where three or more elements join.
Path: A trace of adjoining basic elements with no repition.
Branch: A path connecting two nodes.
Essential Branch: A path connecting two essential nodes.
Loop: First and last node are the same node.
Mesh: A loop that does not enclose another loop.

3.3 Nodal Analysis:

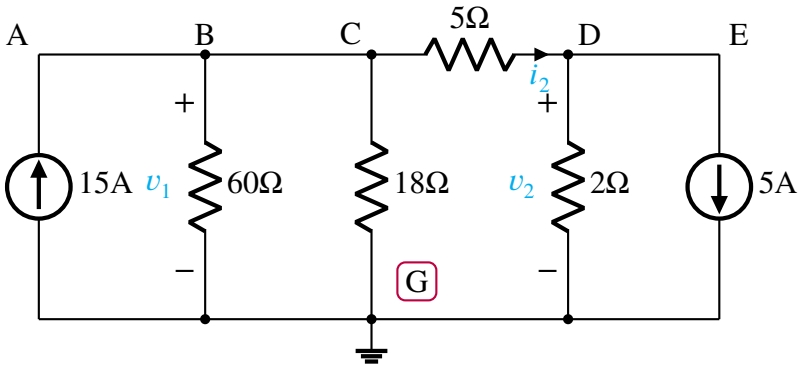
Nodal analysis is known as the node-voltage method. This is a method calculating node voltages in a planar circuit. It utilizes the fundamentals of KCL and Ohm's Law.

3.3.1 Procedure

1. Identify each essential node.
2. Pick and label a reference node (usually ground), then label the node voltages at the remaining essential nodes.
3. Write KCL equations for each non-reference essential node.

4. Solve equations.
5. Find voltages, power and current at each node.

3.3.2 NA Example



Voltage at Node C:

$$\begin{aligned} 15 &= \frac{V_C - 0}{60} + \frac{V_C - 0}{15} + \frac{V_C - V_D}{5} \\ &= \frac{V_C}{60} + \frac{V_C}{15} + \frac{V_C - V_D}{5} \\ 15 &= V_C(\frac{17}{60}) + V_D(-\frac{1}{5}) \end{aligned}$$

Voltage at Node D:

$$\begin{aligned} \frac{V_C - V_D}{5} &= \frac{V_D - 0}{2} + 5 \\ -5 &= \frac{V_D - V_C}{5} + \frac{V_D}{2} \\ -5 &= V_C(-\frac{1}{5}) + V_D(\frac{7}{10}) \end{aligned}$$

Solving with GC:

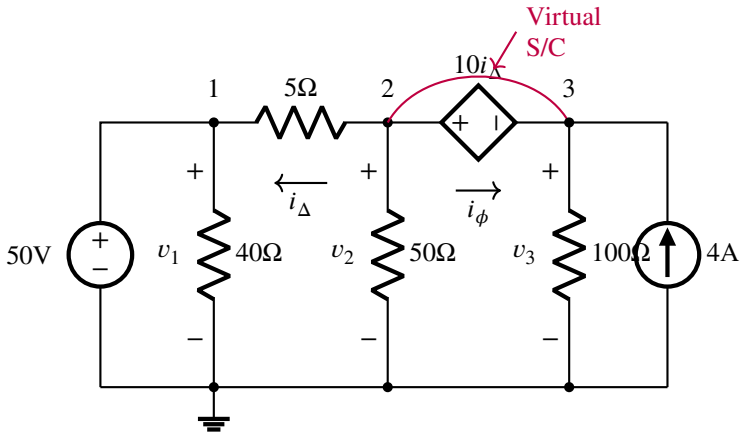
$$V_C = 60V$$

$$V_D = 10V$$

3.4 Super Node:

A circuit with a voltage source that does not have a series resistance that would permit a conversion to a current source. In this situtation there are two options that can be done:

1. Pace a comparitavely small resistor in series to the voltage source.
2. Use the supernode approach; That is to assume a virtual S/C across the voltage source.



At Node 2:

$$\frac{V_3-0}{100}+\frac{V_2-0}{50}+\frac{V_2-V_1}{5}=4$$
$$V_3(\frac{1}{100})+V_2(\frac{11}{50})+V_1(-\frac{1}{5})=4$$

Using KVL:

$$\begin{aligned}V_3 &= V_2+10i_\phi\\ V_1 &= 50V\\ i_\phi &= \frac{V_2-V_1}{5}\\ &= \frac{V_2-50}{5}\\ V_3 &= V_2+10(\frac{V_2-50}{5})\\ &= 3V_2-100\end{aligned}$$

Solving for Node voltages:

$$V_1 = 50V$$
$$V_2 = 60V$$
$$V_3 = 80V$$
$$i_\phi = 2A$$

4 Mesh Analysis

4.1 Branch Current Analysis:

1. Assign a distinct current of arbitrary direction to each direction.
2. Polarities of circuit elements are determined by assumed current direction.
3. Apply KVL around each loop.
4. Apply KCL at each essential node.
5. Solve for current values.

4.2 Mesh Analysis:

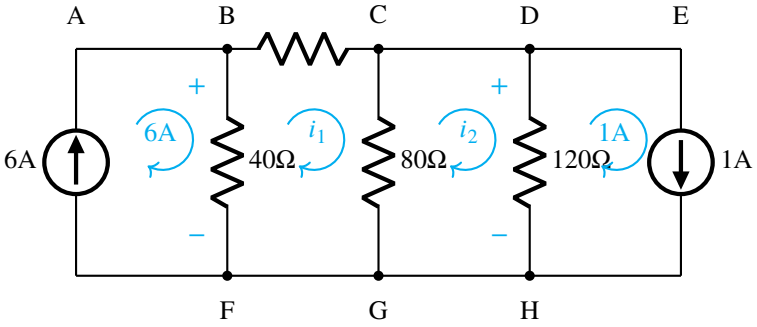
Extension of the above method. It defines currents in such a way that there is no need to apply KCL.

- Define currents for each mush of the networks.
- Branch current is now expressed in terms of unique mesh currents.

4.2.1 Procedure

1. Assign distinct current to each mesh. The easiest way is have them all going in the same direction.
2. Indictate polarities through the loop with respect to current direction.
3. Apply KVL on each mesh.
4. Solve for currents.

4.2.2 Example



Mesh GFBCG:

$$\begin{aligned} 0 &= 40(i_1 - 6) + 8i_1 + 80(i_1 - i_2) \\ 240 &= 128i_1 - 80i_2 \end{aligned}$$

Mesh GCDHG:

$$\begin{aligned} 0 &= 80(i_2 - i_1) + 120(i_2 - 1) \\ 120 &= -80i_1 + 120i_2 \end{aligned}$$

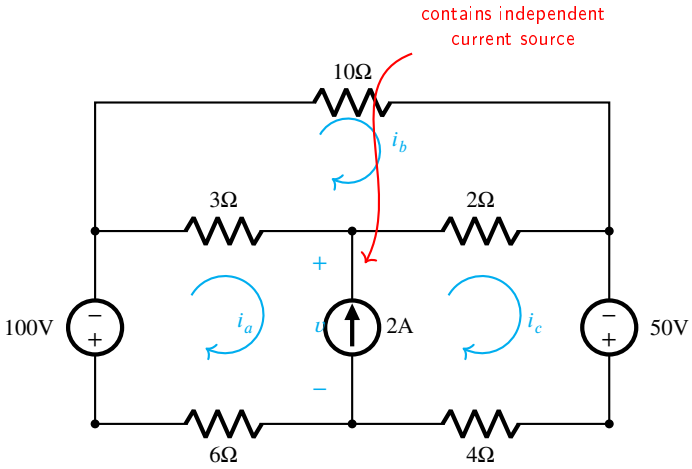
Solving with GC gives:

$$i_1 = 3A$$

$$i_2 = 1.8A$$

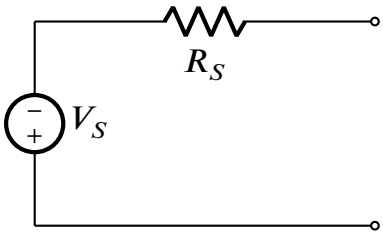
4.3 Super Mesh:

Where an independent current has no parrallel resistor to convert to a voltage source. Assume a virtual O/C and apply the mesh analysis method. Add constraint for KCL at its node.

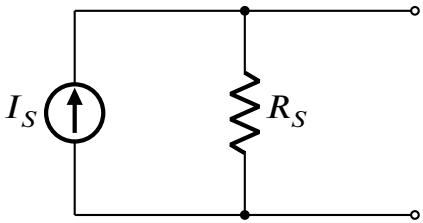


5 Equivalent Circuits

5.1 Source Conversion:



$$V_S = I_S R_S$$

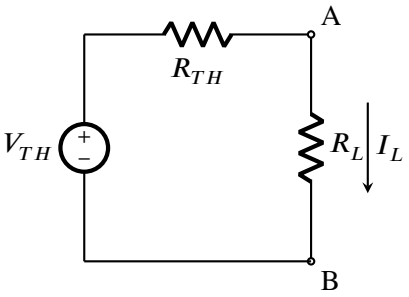


$$I_S = \frac{V_S}{R_S}$$

A voltage source in series with a resistor has an equivalent current in parrallel. There are two main types of equivalent circuits. Thévenin equivalent is with

voltage, where as the Norton equivalent deals with current. Using the rules in the image above, they can be converted between.

5.1.1 Thévenin Circuits



If R_L is infinitely large, causing an O/C $V_{TH} = V_{AB}$. Else if R_L is 0, it causes a S/C, making $R_{TH} = \frac{V_{TH}}{I_L}$.

5.1.2 Norton Circuits

Usualy a Norton equivalent circuit is easier too calculate the S/C current then O/C voltage. However most of the time it is easier to convert from the Thévenin circuit.

5.2 Thévenin Equivalent Resistance:

Only works with independent sources. The short cut method:

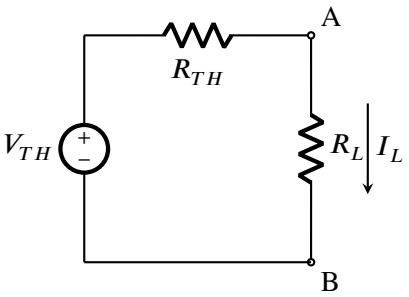
- Deactivate all power sources
  - Ideal voltage sources become S/C
  - Ideal current sources become O/C
  - Internal resistance sources are included in circuit
- Resistance is calculated from the reference terminals

5.3 Maximum Power Transfer:

Important parameters describing the power transfer from source too load. It tells us 2 main things:

1. Amount of power transferred

2. Efficiency



$$P_L = (\frac{V_{TH}}{R_{TH} + R_L})^2 \times R_L$$

Max Power are $R_{TH} = R_L$

$$\therefore \ P_{L_{max}} = \frac{V_{TH}^2}{4R_{TH}}$$

$$\text{Eff} = \frac{P_L}{P_S} = \frac{R_L}{R_{TH}R_L}$$

5.4 Superposition Theorem:

Linear circuit systems observe the principal of superposition. That is the total system response is the sum of all individual responses.

Voltage sources are S/C
Current sources are O/C

Utilized to analyse circuits with both DC & AC components.

6 RC Circuits

6.1 Capacitors:

Constructed using a dialectric material middle, sandwhiched between two conducting plates. As it charges an electric field is generated. As such its true

characteristics are shown when a change in voltage or current take places. To a DC source there is zero change in current because capacitors act an O/C.

$$i = C\frac{dv}{dt}$$

Voltage cannot change instantaneously as it results in infinite induced current. It takes time for voltage to reach desired voltage and vice versa to reach 0V.

$$\begin{aligned}
i &= C\frac{dv}{dt} \\
idt &= Cdv \\
dv &= \frac{1}{C}idt \\
\int_{v(t_0)}^{v(t)} 1dx &= \frac{1}{C}\int_{t_0}^{t} idx \\
v(t) - v(t_0) &= \frac{1}{C}\int_{t_0}^{t} idx \\
v(t) &= \frac{1}{C}\int_{t_0}^{t} idx + v(t_0)
\end{aligned}$$

$$\begin{aligned}
p &= vi \\
&= vC\frac{dv}{dt} \\
&= i[\frac{1}{C}\int_{t_0}^{t} idx + v(t_0)] \\
W &= \frac{Cv^2}{2}
\end{aligned}$$

Power is in Watts and Energy (W) is in Joules.

6.2 Equivalent Capacitence:

6.2.1 Series

$$C_{eq} = \frac{1}{\sum_{n=1}^{k} \frac{1}{C_n}}$$

$$v(t_0) = v_1(t_0) + v_2(t_0) + ... + v_n(t_0)$$

6.2.2 Parallel

$$C_{eq} = \sum_{n=1}^{k} C_n$$

6.3 Inductors:

Inductors can be thought of as the dual to the capacitor. Similar to the capacitor, inductors only exhibit true behaviour on current and voltage changes. Inductors are created by wrapping wire material around a ferromagnetic core. As such they are designed to generate magnetic field linking the unit. Inductors need a change in current to induce voltage. A constant current (DC) would induce OV.

$$\begin{aligned} v &= L\frac{di}{dt} \\ vdt &= Ldi \\ L\int_{i(t_0)}^{i(t)} 1dx &= \int_{t_0}^{t} vdy \\ i(t) - i(t_0) &= \frac{1}{L}\int_{t_0}^{t} vdy \\ i(t) &= \frac{1}{L}\int_{t_0}^{t} vdy + i(t_0) \end{aligned}$$

$$\begin{aligned} p &= vi \\ &= iL\frac{di}{dt} \\ &= v[\frac{1}{L}\int_{t_0}^{t} vdy + i(t_0)] \\ w &= \frac{Li^2}{2} \end{aligned}$$

6.4 Equivalent Inductance:

6.4.1 Series

$$L_{eq} = \sum_{n=1}^{k} L_n$$

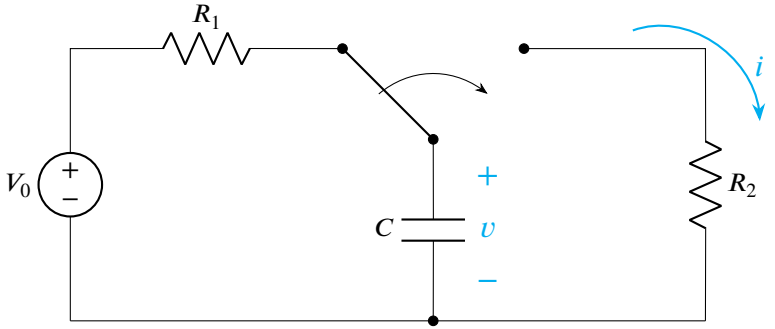
6.4.2 Parallel

$$L_{eq} = \frac{1}{\sum_{n=1}^{k} \frac{1}{L_n}}$$

$$i(t_0) = i_1(t_0) + i_2(t_0) + ... + i_n(t_0)$$

6.5 Discharging Circuit:

When voltage is applied across a capacitor it charges. Once the power is disconnected the capacitor discharges. This is called the natural response.



Assume that the switch as been in first position for a long time. Thus the $v_c(t_0^-) = V_0$.

$$\therefore v_c(t_0^+) = V_0$$

Using KCL:

$$\begin{aligned} i + i_C &= 0 \\ \frac{v}{R} + C\frac{dv}{dt} &= 0 \\ v(t) &= \frac{\int e^{\int \frac{1}{RC} dt} \times 0 dt + K}{e^{\int \frac{1}{RC} dt}} \\ &= K e^{-\frac{t}{RC}} \\ i(t) &= (\frac{V_0}{R}) e^{-\frac{t}{RC}} \end{aligned}$$

Power disssapated

$$\begin{aligned} p(t) =& i(t) \times v(t) \\ =& \frac{V_0^2 e^{-\frac{2t}{RC}}}{R}, \ t > 0 \end{aligned}$$

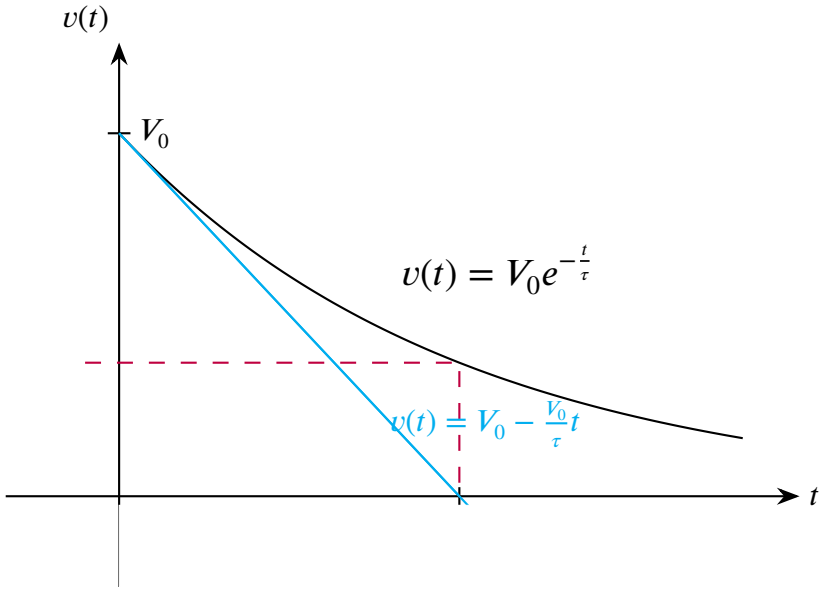
Energy Delivered

$$W = \frac{1}{2} C V_0^2 [1 - e^{-1 \frac{2t}{RC}}], \ t \geq 0$$

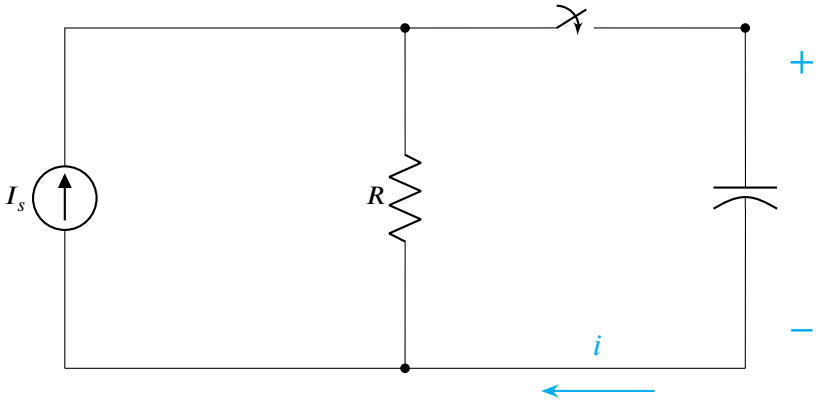
6.6 RC Constant:

Since RC are known values, and they determine v(t) and i(t), they are the time constant (in seconds).

$$\tau = RC$$



6.7 Charging through a Resistor:



$$C\frac{dv}{dt} + \frac{V}{R} = I_S$$

$$v(t) = \frac{\int e^{\int \frac{1}{RC}dt} \times \frac{I_S}{C}dt + K}{e^{\int \frac{1}{RC}}}$$

$$= RI_S + Ke^{-\frac{t}{RC}}$$

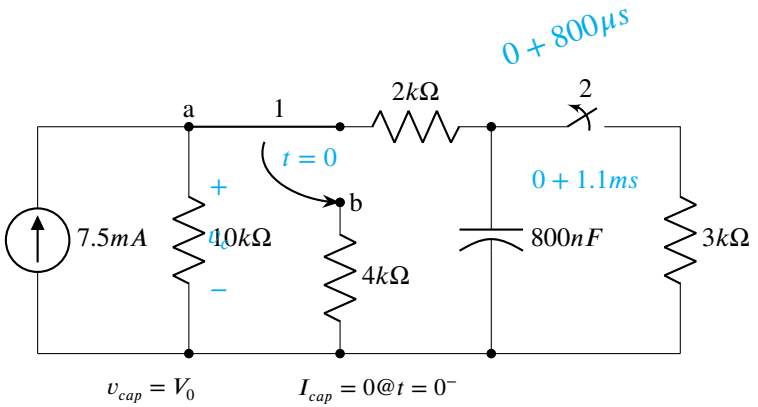
Assuming at $t = 0$ $v = V_0$

$$K = V_0 - RI_S$$

$$v(t) = RI_S + (V_0 - RI_S)e^{-\frac{t}{RC}},\ t \geq 0$$

$$i(t) = I_S e^{-\frac{t}{RC}},\ t > 0$$

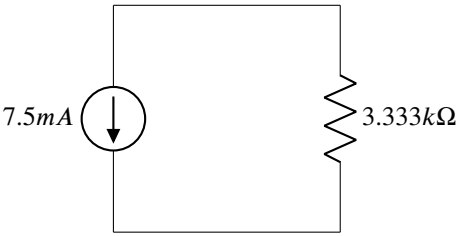
6.8 Exmaple:



Switch 1 has been in position 'a' and switch 2 has been closed for a long time.
At t = 0 switch 1 moves instantaneously to position 'b'.
800 μs later, switch 2 opens and then recloses after another 300 μs.
Find v_o at t =1.5ms

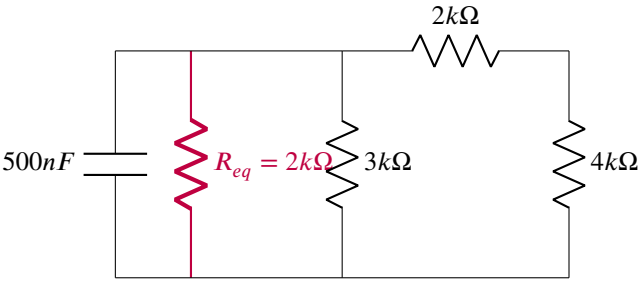
Time	SW1	SW2
0^-	a	Closed
0^+	b	Closed
$800\mu s$	b	Open

$@\ t = 0^-$



$$
\begin{aligned}
v(0^+) &= I_S R_{eq} \\
&= 7.5 \times 10^{-3} \times 3.3333 \times 10^3 \\
&= -25V \\
v_{cap}(0^-) &= -25 \times \frac{3}{2+5} \\
&= -15V
\end{aligned}
$$

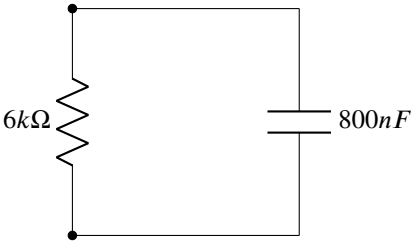
@ $t = 0^+$



At $v(0^+) = v(0^-) = -15V$

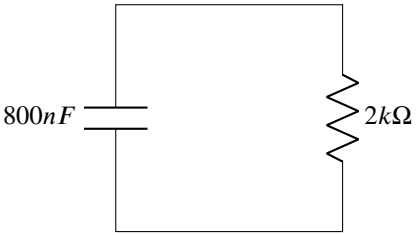
$$
\begin{aligned}
v(t) &= V_0 e^{-\frac{t}{RC}} \\
&= -15e^{-\frac{t}{2\times10^3\times500\times10^{-9}}} \\
&= -15e^{-10^3 t},\ 0^+ \le t \le 800^-\mu s
\end{aligned}
$$

@ $t = 800\mu s$



$$
\begin{aligned}
v(800^{-}\mu s) &= -15e^{-10^3\times 800\times 10^{-6}} \\
&= -6.7399V \\
\tau_{800\mu s^{+}} &= RC \\
&= 6\times 10^3 \times 500 \times 10^{-9} \\
&= 3\times 10^{-3} \\
v(t) &= -6.74e^{-\frac{t-800\times 10^{-6}}{3\times 10^3}}, 800\mu s^{+} \leq t \leq 1100\mu s^{-}
\end{aligned}
$$

$$@\ t = 1100\mu s$$



$$
\begin{aligned}
v(1100\mu s^{-}) &= -6.1V \\
\tau_{1100\mu s^{+}} &= RC \\
&= 2\times 10^3 \times 500 \times 10^{-9} \\
&= 10^{-3}
\end{aligned}
$$

$$
\begin{aligned}
v(t) &= -6.1e^{-\frac{t-1100\times 10^{-6}}{10^{-3}}} \\
v(1500\mu s) &\approx -4.1V
\end{aligned}
$$

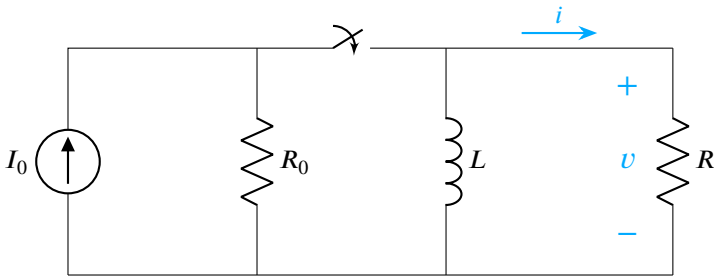
7 RL Circuits

7.1 Energy Flow in Inductors:

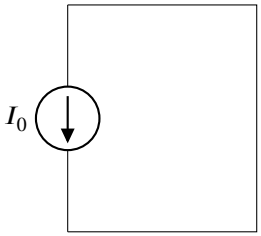
Here, only circuits consisting of inductors and resistors will be considered. This simplifies equations to first ODEs. Inductors are similar to capacitors in that it stores energy.

7.2 Steady State:

When an inductor reaches DC steady state, it acts as a S/C.



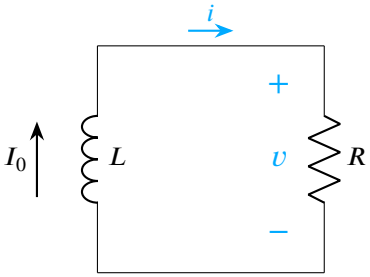
At DC Steady State:



$$i(0^-) = I_0$$

$$i(0^+) = I_0$$

After Switching



$$
\begin{aligned}
L\frac{di}{dt} + iR &= 0 \\
\frac{di}{dt} + \frac{iR}{L} &= 0 \\
i(t) &= \frac{\int e^{\int \frac{R}{L}dt} \times 0 dt + K}{e^{\int \frac{R}{L}dt}} \\
&= Ke^{-\frac{Rt}{L}}
\end{aligned}
$$

$$
\tau = \frac{L}{R}
$$

$$
\begin{aligned}
\therefore i(t) &= Ke^{-\frac{t}{\tau_L}} \\
&= I_0 e^{-\frac{t}{\tau_L}}
\end{aligned}
$$

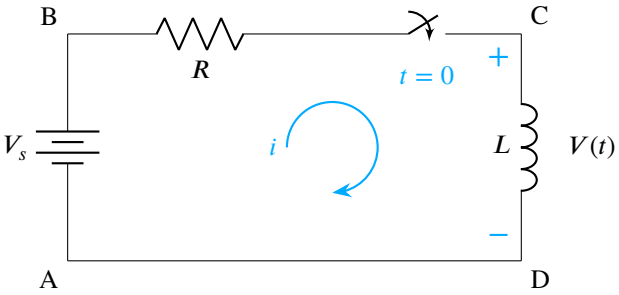
$$
\begin{aligned}
v_L(t) &= L\frac{di}{dt} \\
&= L(-I_0 \cdot \frac{1}{\tau_L} e^{-\frac{t}{\tau_L}} \\
&= -I_0 R e^{-\frac{t}{\tau_L}} \\
v_R(t) &= -v_L(t) \\
&= I_0 R e^{-\frac{t}{\tau_L}}
\end{aligned}
$$

7.3 Power & Energy in Inductors:

$$\begin{aligned} p(t) =& i(t) \times v(t) \\ =& I_0 e^{-\frac{t}{\tau_L}} \cdot I_0 R e^{-\frac{t}{\tau_L}} \\ =& I_0^2 R e^{-\frac{2t}{\tau_L}}, \ t > 0 \end{aligned}$$

$$\begin{aligned} w =& \int_0^t p(x) dx \\ =& \int_0^t I_0 R e^{-\frac{2x}{\tau_L}} dx \\ =& \frac{1}{2} L I_0^2 [1 - e^{-\frac{2t}{\tau_L}}], \ t \geq 0 \end{aligned}$$

7.4 Get Energy going through Resistor:



KVL around ABCDA

$$\begin{aligned} v_R + v_L =& v_S \\ iR + L\frac{dv}{dt} =& v_S \\ \frac{di}{dt} + \frac{iR}{L} =& \frac{v_S}{L} \\ i(t) =& \frac{\int e^{\int \frac{R}{L} dt} (\frac{v_S}{L}) dt + K}{e^{\int \frac{R}{L} dt}} \\ =& \frac{v_S}{R} + K e^{-\frac{t}{\tau_L}} \end{aligned}$$

Assuming @ $t = 0$, $i = I_0$, that when the switch closes there is some energy already in the inductor.

$$K = I_0 - \frac{I_S}{R}$$
$$\therefore i(t) = \frac{v_S}{R} + (I_0 - \frac{v_S}{R})e^{-\frac{t}{\tau_L}}$$

7.5 Summary:

If at $t = 0$, $I_0 = 0$ then:

$$i(t) = \frac{v_S}{R}(1 - e^{-\frac{t}{\tau_L}})$$
$$v_L(t) = v_S e^{-\frac{t}{\tau_L}},\ \tau_L = \frac{L}{R}$$
$$v_R = V_S - v_L$$

8 RLC Circuits - Parallel

8.1 Second Order Differential Equations:

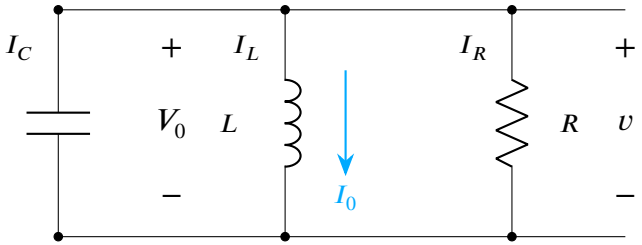
RLC circuits consist of 2nd ODEs in homogeneous constant coefficient equations.

$$\frac{d^2y}{dx^2} + A\frac{dy}{dx} + By = 0$$
$$s^2 + As + B = 0$$
$$s = \frac{-A \pm \sqrt{A^2 - 4B}}{2}$$
$$\therefore y = k_1 e^{s_1 t} + k_2 e^{s_1 t}$$

k_1 & k_2 are found in the initial conditions within the circuit.

8.2 Natural Response:

Discharging within a circuit,



$$
\begin{aligned}
I_C + I_L + I_R + I_0 &= 0 \\
C\frac{dv}{dt} + \frac{1}{L}\int_0^t v\,d\tau + \frac{V}{R} + I_0 &= 0 \\
\frac{dv}{dt} + \frac{1}{LC}\int_0^t v\,d\tau + \frac{V}{RC} + \frac{I_0}{C} &= 0 \\
\frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{1}{LC}V &= 0
\end{aligned}
$$

$$
\begin{aligned}
v(t) &= k_1 e^{s_1 t} + k_2 e^{s_2 t} \\
s_1 &= \frac{-1}{2RC} + \sqrt{(\frac{1}{2RC})^2 - \frac{1}{LC}} \\
s_2 &= \frac{-1}{2RC} - \sqrt{(\frac{1}{2RC})^2 - \frac{1}{LC}}
\end{aligned}
$$

Neper Frequency of a parallel RLC circuit:

$$\alpha = \frac{1}{2RC}$$

Resonant Radian Frequency of paralllel RLC circuit:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

All of these are frequencies and measured in $\frac{\text{rad}}{\text{s}}$.

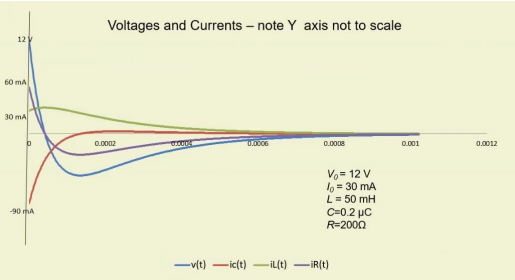
Iff	Then
$\alpha^2 - \omega_0^2 > 0$	Over Damped
$\alpha^2 - \omega_0^2 < 0$	Under Damped
$\alpha^2 - \omega_0^2 = 0$	Critically Damped

Using I_0 throug inductor and V_0 across capacitor.

$$V_0 = k_1 + k_2$$

$$C(k_1 s_1 + k_2 s_2) = -I_0 + \frac{V_0}{R}$$

8.3 Over-damped Response:



8.4 Under-damped:

$$(\frac{1}{2RC})^2 - \frac{1}{LC} < 0$$

$$\begin{aligned} s &= -\alpha + \sqrt{\alpha^2 - \omega_0^2} \\ &= -\alpha + j\sqrt{\omega_0^2 - \alpha^2} \end{aligned}$$

$$\begin{aligned} s_1 &= -\alpha + j\omega_d \\ s_2 &= -\alpha - j\omega_d \end{aligned}$$

This is where ω_d is the damped radian frequency.

$$v(t) = k_1 e^{-\alpha t}\cos(\omega_d t) + k_2 e^{-\alpha t}\sin(\omega_d t)$$

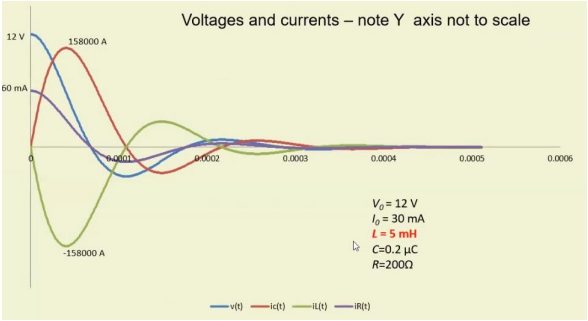
$$V_0 = k_1$$

$$C(-k_1\alpha + k_2\omega_d) = -(I_0 + \frac{V_0}{R})$$

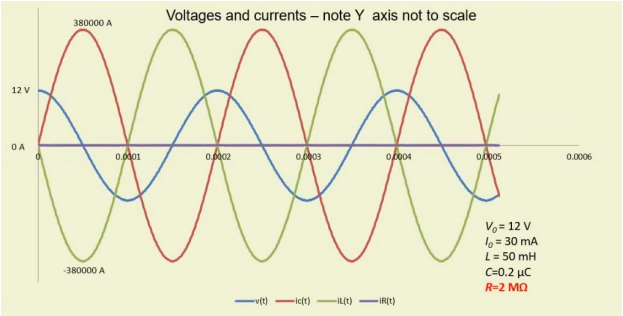
- Oscillatory with period damped radian frequency.
- Amplitude decreases at the rate α, this is called the damping factor.

- Oscillates due to two stored energies.
- Overdamped - Takes times to reach final value, but transients do not exceeed intitial value.
- Underdamped - Reaches final value, oscilates around it before reaching steady state. Transient spikes can be few orders of magnitude larger then initial values.

8.4.1 Small Resistor



8.4.2 Large Resistor



8.5 Critically Damped Response:

$$(\frac{1}{RC})^2 - \frac{1}{LC} = 0$$

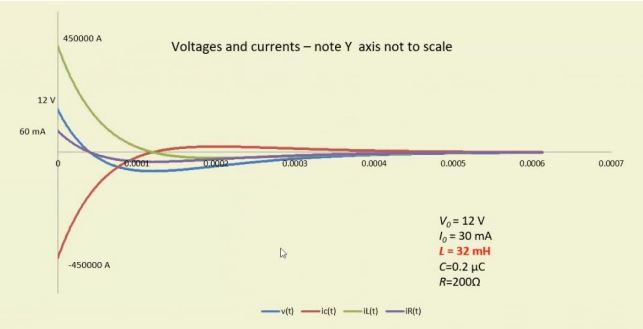
$$\alpha = \omega_0$$

$$v(t) = k_1 t e^{-\alpha t} + k_2 e^{-\alpha t}$$

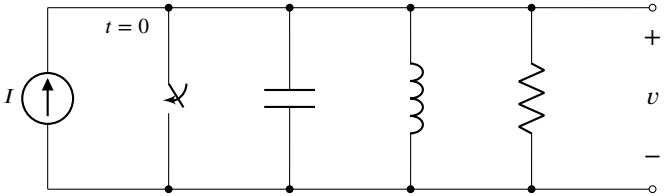
$$V_0 = k_2$$

$$C(k_1 - k_2\alpha) = -(I_0 + \frac{V_0}{R})$$

Rarely happens as values need to match exactly for it to be achieved.



8.6 Step Response:



$$I_L + I_R + I_C = I$$

$$i_L + \frac{V}{R} + C\frac{dv}{dt} = I$$

$$i_L + \frac{L\frac{di_L}{dt}}{R} + C\frac{d(\frac{di_L}{dt})}{dt} = I$$

$$i_L + \frac{L}{R}\frac{di_L}{dt} + C\frac{d^2i_L}{dt^2} = I$$

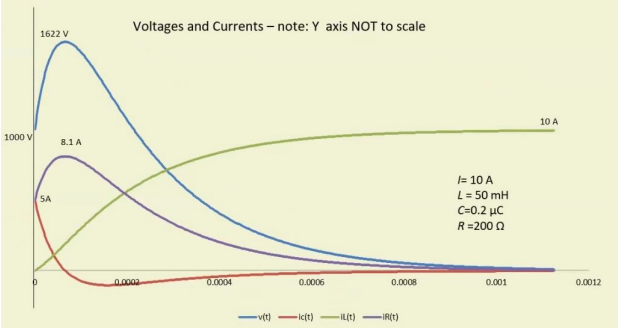
$$\frac{d^2i_L}{dt^2} + \frac{1}{RC}\frac{di_L}{dt} + \frac{1}{LC}i_L = \frac{I}{LC}$$

Non-homogeneous equations:

$$i_L = I_f + I_H$$

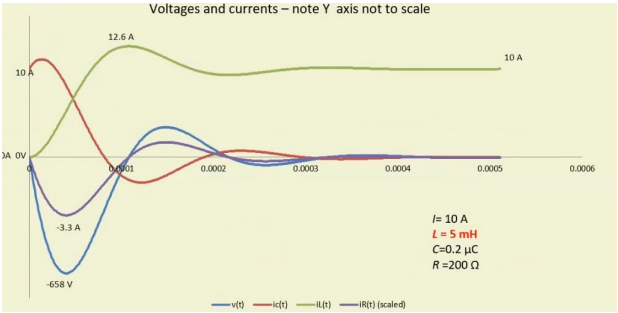
8.6.1 Over-damped

$$i_L(t) = I_f + k_1 e^{s_1 t} + k_2 e^{s_2 t}$$



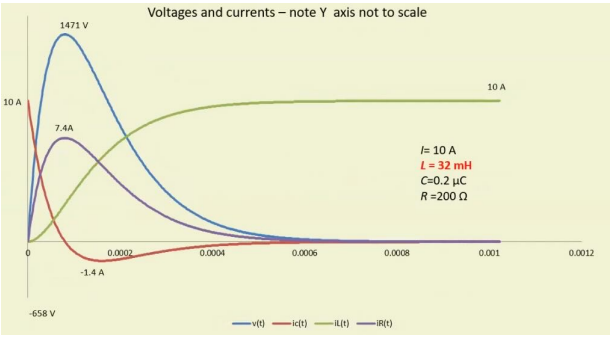
8.6.2 Under-damped

$$i_L(t) = I_f + k_1 e^{-\alpha t} \cos(\omega_d t) + k_2 e^{-\alpha t} \sin(\omega_d t)$$



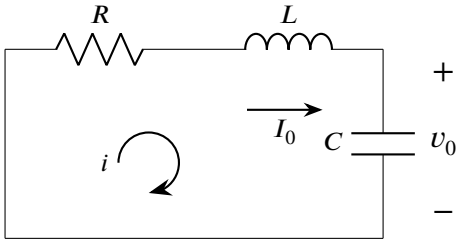
8.6.3 Critically-damped

$$i_L(t) = I_f + k_1 t e^{-\alpha t} + k_2 e^{-\alpha t}$$



9 RLC Circuit - Series

9.1 Natural Response of RLC Series Circuits:



Energy stored in both capacitor and inductor.

$$V_R + V_L + V_C = 0$$

$$iR + L\frac{di}{dt} + \frac{1}{C}\int_0^t v\, d\tau + V_0 = 0$$

$$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{1}{LC}i = 0$$

$$s_1 = -\frac{R}{2L} + \sqrt{(\frac{R}{2L})^2 - \frac{1}{LC}}$$

$$s_2 = -\frac{R}{2L} - \sqrt{(\frac{R}{2L})^2 - \frac{1}{LC}}$$

Neper Frequency Series:

$$\alpha = \frac{R}{2L}$$

Resonant Radian Frequency:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

9.1.1 Over-damped

$$i(t) = k_1 e^{s_1 t} + k_2 e^{s_2 t}$$

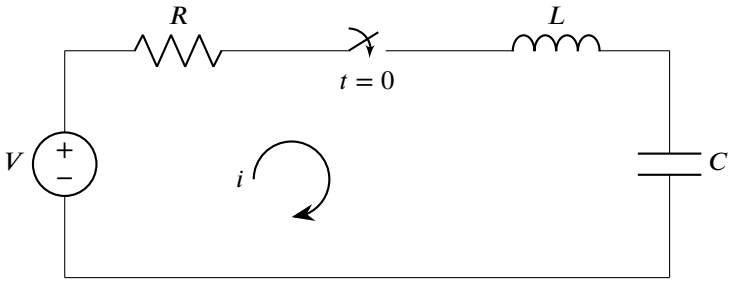
9.1.2 Under-damped

$$i(t) = k_1 e^{-\alpha t} \cos(\omega_d t) + k_2 e^{-\alpha t} \sin(\omega_d t)$$

9.1.3 Critically-damped

$$i(t) = k_1 t e^{-\alpha t} + k_2 e^{-\alpha t}$$

9.2 Step Response:



$$\begin{aligned} V_L + V_C + V_R &= V \\ L\frac{di}{dt} + V_C + iR &= V \\ \frac{d^2 V_C}{dt^2} + \frac{R}{L}\frac{dV_C}{dt} + \frac{1}{LC}V_C &= \frac{V}{LC} \end{aligned}$$

9.2.1 Over-damped

$$v_C(t) = V_f + k_1 e^{s_1 t} + k_2 e^{s_2 t}$$

9.2.2 Under-damped

$$v_C(t) = V_f + k_1 e^{-\alpha t} \cos(\omega_d t) + k_2 e^{-\alpha t} \sin(\omega_d t)$$

9.2.3 Critically-damped

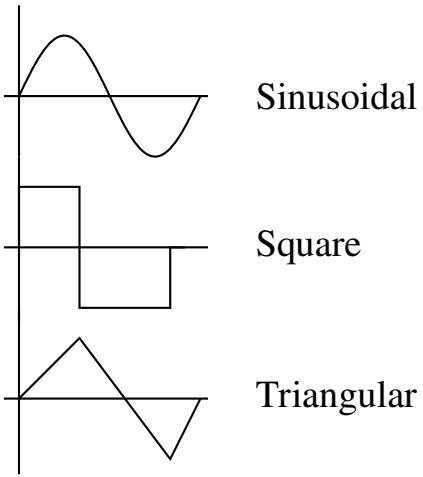
$$v_C(t) = V_f + k_1 t e^{-\alpha t} + k_2 e^{-\alpha t}$$

10 Steady State Sinusodial Analysis - Part I

Most of the theorems within DC steady state will works within AC analysis. For example, Norton and Thévenin equivalents, KCL and KVL as well as mesh and nodal calculations.

10.1 Introduction:

Most power sources generate periodic, time varying voltage and current.

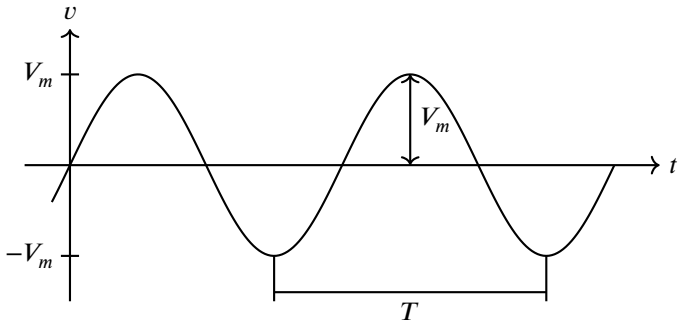


10.2 Definition:

- Waveform - The path traced by quantity
- Instant value - Magnitude of that time point
- Peak Amplitude - Maximum value measured from average
- Peak Value - Max value measured from 0
- P-P value - Full value difference between Max and Min values
- Periodic waveform - Continually repeats itself after set time

- Period - Time of the period waveform
- Cycle - The portion contained in one period of time
- Frequency - The number of cycles per second

10.3 Sinusoidal Waveform:



$$\begin{aligned} v(t) &= V_m \cos(\omega t + \phi) \\ &= V_m \sin(\omega t + \phi) \end{aligned}$$

Where:

- V_m - Peak amplitude (V)
- T - Period (s)
- $f = \frac{1}{T}$ - Frequency (Hz)
- ω - $2\pi f$ (rad/s)
- ϕ - Phase angle (rad)

The ϕ defines the start of $v(t)$ or $i(t)$ at $t = 0$. It is important to make sure ω and ϕ are in the same units. A positive value for ϕ shifts the waveform left, with a negative ϕ value shifting it right. However, ϕ is usually given in degrees, rather than rad.

Angular Velocity:

$$\begin{aligned}\omega &= \frac{\text{Dist}(\alpha)}{\text{time}(t)} \\ \alpha &= \omega t \\ \omega &= 2\pi f \\ &= \frac{2\pi}{T}\end{aligned}$$

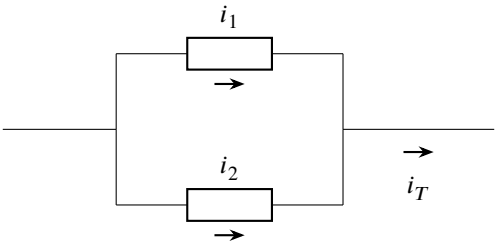
10.4 RMS Value:

It is the root of the mean value of the squared funciont over 1 period.

$$\begin{aligned}V_{RMS} &= \sqrt{\frac{1}{T}\int_{t_0}^{t_0+T} V_m^2 \cos^2(\omega t + \phi)dt} \\ &= \frac{V_m}{\sqrt{2}}\end{aligned}$$

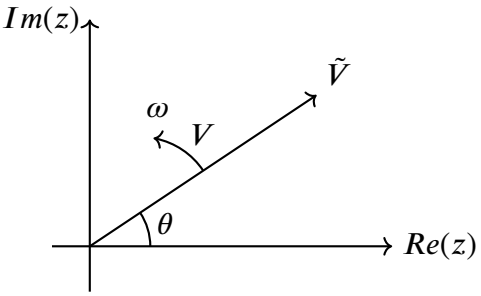
The short cut method is only applicable to sinusodial waveform.

10.5 AC Analysis:



$$i_1 = 5\sin(\omega t + 30°)$$
$$i_2 = 6\sin(\omega t + 60°)$$
$$i_T = i_1 + i_2$$

10.6 Phasors:



10.7 Phasors:

Euler's Identity:

$$e^{\pm j\theta} = \cos(\theta) \pm j\sin(\theta)$$
$$\cos(\theta) = R\{e^{j\theta}\}$$
$$\sin(\theta) = I\{e^{j\theta}\}$$

$$\begin{aligned} V &= V_m \cos(\omega t + \phi) \\ &= V_m R\{e^{j(\omega t+\phi)}\} \\ &= R\{V_m e^{j\theta} e^{j\omega t}\} \\ V &= V_m e^{j\theta} \\ &= V_m \cos(\phi) + jV_m \sin(\phi) \\ &= P\{V_m \cos(\omega t + \phi)\} \end{aligned}$$

$$V_m \cos(\omega t + \phi) \Rightarrow V_m e^{j\theta} \equiv V_m \angle\phi^\circ$$

Inversed:

$$\begin{aligned} P^{-1}\{V_m\angle\phi\} &\equiv P^{-1}\{V_m e^{j\theta}\} \\ &= R\{V_m e^{j\omega t} e^{j\theta}\} \\ &\Rightarrow V_m \cos(\omega t + \phi) \end{aligned}$$

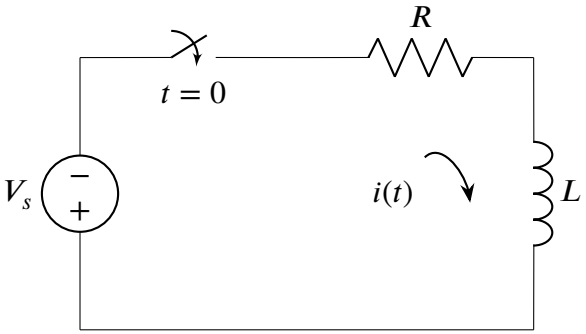
10.8 AC Analysis Using Phasors:

- All quantities must be in same frequency.
- Do no care about the transient effects.

- The steady state solution is $R\{Ae^{j\beta}e^{j\omega t}\}$. This only is true in linear circuits.

10.9 Phasor Analysis:

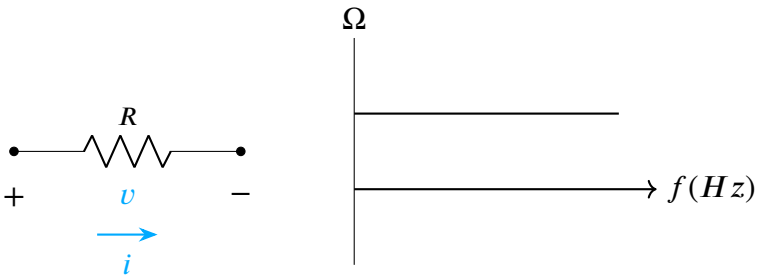
Find v_s when $i = I_m \cos(\omega t + \phi)$



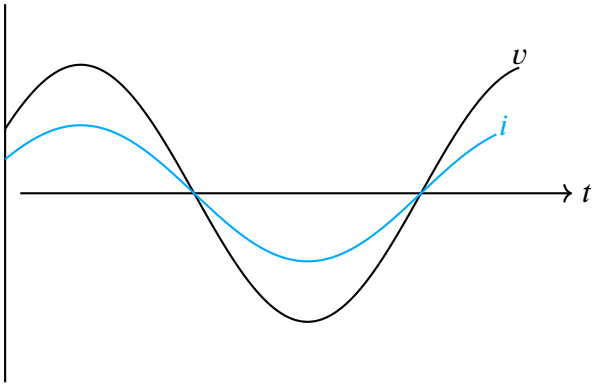
$$
\begin{aligned}
v_s &= iR + L\frac{di}{dt} \\
R\{V_m e^{j\beta}e^{j\omega t}\} &= R\{RI_m e^{j\omega t}e^{j\phi}\} + R\{j\omega L I_m e^{j\omega t}e^{j\phi}\} \\
&= R\{(RI_m e^{j\phi} + j\omega L I_m e^{j\phi})e^{j\omega t}\} \\
V_m e^{j\beta} &= RI_m e^{j\phi} + j\omega L I_m e^{j\phi}
\end{aligned}
$$

10.10 Phasor Domains:

10.10.1 Resistor

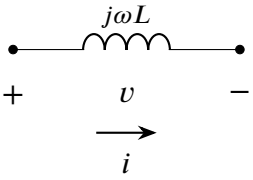


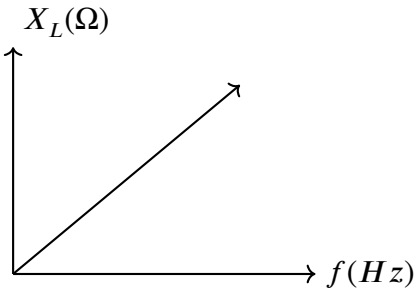
$$
\begin{aligned}
i &= I_m \cos(\omega t + \phi) \\
v &= iR \\
&= RI_m \cos(\omega t + \phi) \\
&= RI_m e^{j\theta} \\
&= RI_m \angle\theta
\end{aligned}
$$



Voltage and Current in Phase.

10.10.2 Inductor

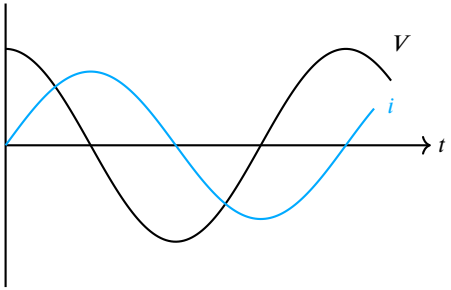




$f = 0Hz \quad L \text{ is } S/C$

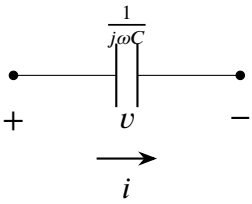
$f = \infty Hz \quad L \text{ is } O/C$

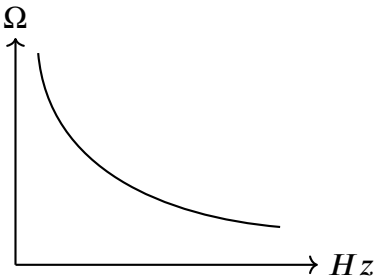
$$
\begin{aligned}
i &= I_m \cos(\omega t + \phi) \\
v &= L\frac{di}{dt} \\
&= -\omega L I_m \sin(\omega t + \phi) \\
&= -\omega L I_m \cos(\omega t + \phi - 90°) \\
&= -\omega L I_m e^{j\omega t} e^{j\theta} e^{j+90°} \\
&= j\omega L I_m e^{j\phi} \\
&= j\omega L I_m \angle\phi
\end{aligned}
$$



Current lags the Voltage by 90°

10.10.3 Capacitor

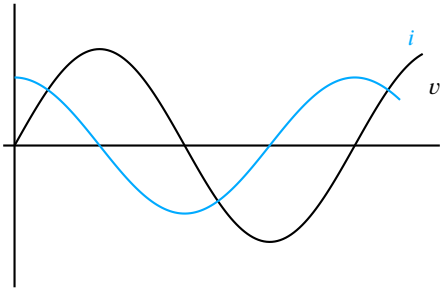




$f = 0Hz$ Cap is O/C

$f = \infty Hz$ Cap is S/C

$$
\begin{aligned}
v &= V_m \cos(\omega t + \phi) \\
i &= C\frac{dv}{dt} \\
&= -\omega C V_m \sin(\omega t + \phi) \\
&= -\omega C V_m \cos(\omega t + \phi - 90°) \\
&= j\omega C V_m e^{j\phi} \\
&= j\omega C V_m \angle\phi
\end{aligned}
$$



Current leads Voltage by 90°

10.11 Impedance:

Basically resistance within AC circuits.

$$v = Zi$$

- Resistor - $Z_R = R = R\angle 0°$
- Inductor - $Z_L = j\omega L = \omega L\angle 90° = 2\pi f L\angle 90°$
- Capacitor - $Z_C = \frac{-j}{\omega C} = \frac{1}{\omega C}\angle -90° = \frac{1}{2\pi f C}\angle -90°$

Although Z can be a complex number, it is NOT a phasor. Resistance is the real part, whereas reactance is the imaginary part. Admittance is the inverse of impedance:

$$Y = \frac{1}{Z}$$

10.12 Equivalent Impedance:

10.12.1 Series

$$Z_{eq} = \sum_{n=1}^{k} Z_n$$

10.12.2 Parallel

$$Z_{eq} = \frac{1}{\sum_{n=1}^{k} \frac{1}{Z_n}}$$

11 AC Power Factor

11.1 Instaneous Power:

Where:

$$\begin{aligned} p &= vi \\ v &= V_m \cos(\omega t + \theta_v) \\ i &= I_m \cos(\omega t + \theta_i) \end{aligned}$$

$$\begin{aligned} v &= V_m \cos(\omega t + \theta_v - \theta_i) \\ i &= I_m \cos(\omega t) \end{aligned}$$

$$\begin{aligned} \therefore p &= V_m \cos(\omega t + \theta_v - \theta_i) \times I_m \cos(\omega t) \\ &= \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) \cos(2\omega t) \\ &\quad - \frac{V_m I_m}{2} \sin(\theta_v - \theta_i) \sin(2\omega t) \end{aligned}$$

It is important to note that the frequency of power is double that of voltage and current. Also at certain points power is a negative value.

11.2 Average Power:

$$\begin{aligned} p &= \frac{1}{T} \int_{t_0}^{t_0+T} vi \, dt \\ &= \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) \end{aligned}$$

This is also called real power and is measured in Watts (W).

11.3 Reactive Power:

$$\begin{aligned} Q =& \frac{V_m I_m}{2} \sin(\theta_v - \theta_i) \\ =& |S| \sqrt{1 - pf^2} \end{aligned}$$

Called reactive power and measured in Volt-Amperes-Reactive (VAR).

11.4 Pure Values:

11.4.1 Resistor

For resistor voltage and current are in the same phase.

$$\therefore \theta_v = \theta_i$$

$$P = \frac{V_m I_m}{2}, \quad Q = 0$$

$$p = P(1 + \cos(2\omega t)$$

Power is always a positive value, this means it always dissapates.

11.4.2 Inductor

Current lags voltage by 90°

$$\therefore \theta_v - \theta_i = 90°$$

$$P = 0 \quad Q = \frac{V_m I_m}{2}$$

$$p = -Q \sin(2\omega t)$$

The average power is 0. Thus, in the positive cycle, energy is stored in the inductor, with the negative cycle, energy is released from the indcutor.

11.4.3 Capacitor

Current leads voltage by 90°

$$\therefore \theta_v - \theta_i = -90°$$

$$P = 0 \quad Q = -\frac{V_m I_m}{2}$$

$$p = Q \sin(2\omega t)$$

Same cycle types as the inductor by reversed. Basically, when the capacitor releases the inductor absorbs, and the capacitor absorbs the inductor releases.

11.5 Power Factor:

$$\cos(\theta_v - \theta_i)$$

Known as the power fator, the angle describes the angle
Lagging pf: $\theta_v - \theta_i > 0$
Leading pf: $\theta_v - \theta_i < 0$

11.6 RMS Value:

This is mostly known as the effective value.

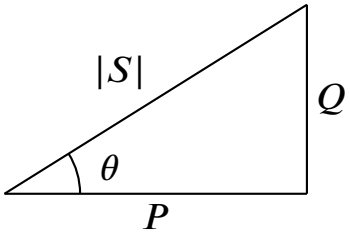
$$\begin{aligned} P =& V_{RMS} I_{RMS} \cos(\theta_v - \theta_i) \\ =& V_{eff} I_{eff} \cos(\theta_v - \theta_i) \end{aligned}$$

$$Q = V_{eff} I_{eff} \sin(\theta_v - \theta_i)$$

11.7 Complex Power:

$$S = P + jQ$$

Complext power as units Volt-Amperes (VA) and is dimensionally akin to the average power. The $|s|$ value is the known as apparent power. It has the same units as complext power.



$$
\begin{aligned}
S =& P + jQ \\
=& V_{RMS}\overline{I_{RMS}} \\
=& \frac{1}{2} V \bar{I}
\end{aligned}
$$

$$
\begin{aligned}
P =& |I_{RMS}|^2 R \\
=& \frac{1}{2} I_m^2 R
\end{aligned}
$$

$$
\begin{aligned}
Q =& |I_{RMS}|^2 X \\
=& \frac{1}{2} I_m^2 X
\end{aligned}
$$

11.8 PF Correction:

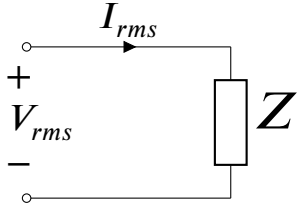
Power dissapated in transmission lines depends on current. Hence, current through the lines should be the smallest values possible. S should be as close to 0 as possible.

$$P = S\cos(\theta)$$

$$pF = \cos(\theta) = \frac{P}{S}$$

$$\theta = \arccos(\frac{P}{S})$$

11.9 Maximum Power Transfer:



$$Z_L = R_L + jX_L \quad \& \quad Z_{TH} = R_{TH} + jX_{TH}$$

$$P = |I_{RMS}|^2 R_L = \frac{|V_{RMS}|^2 R_L}{(R_L + R_{TH})^2 + (X_L + X_{TH})^2}$$

Max power is when:

$$\frac{\partial P}{\partial R_L} = 0 \ \ \& \ \ \frac{\partial P}{\partial X_L} = 0$$

$$Z_L = R_{TH} - jX_{TH} = \overline{Z_{TH}}$$